

AIMO Proof Pilot — Final Grade

Problem (#_ID): 2_ALTORO

Submission: model_1

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

The solution correctly reduces the target ratio to $PQ/PE = HQ/QH'$, where H' is the reflection of H in AC , and identifies that H' lies on the circumcircle — which reaches the partial-progress milestone. However, it then simply asserts a value for this ratio (yielding $a + b = 19$) with no computation, and that value is wrong (the correct ratio is $125/91$, giving $a + b = 216$). The constructive reduction is achieved, but the decisive evaluation is both missing and incorrect, so it scores **1**.